

# Syntactic mismatch and semantic mismatch

Conditionals

30 November 2017

## 1. Recap: putative counterexamples to Modus Ponens

- (i) The paradox of gentle murder (Forrester): A: 'If John kills his mother, he ought to kill her gently'. B: 'John will kill his mother'. C: 'John ought to kill his mother gently'.
- (ii) (A slight variant:) A 'If we're going to leave tomorrow we should start packing now'; B: 'We're going to leave tomorrow'; C: 'We should start packing now'.
- (iii) (McFarlane and Kolodny): A: 'If the miners are in shaft 1 we should flood shaft 2'; B: 'The miners are in shaft 1'; C: 'We should flood shaft 2'.
- (iv) (Yalcin): A: 'If the marble is big, it's likely red'; B: 'The marble is big'; C: 'The marble is likely red'.
- (v) (A slight variant): A: 'If it's Tuesday we must be in Paris'; B: 'It's Tuesday'; C: 'We must be in Paris'.
- (vi) (A different kind of variant): A: 'If the marble is big, I'm confident that it's red'; B: 'The marble is big'; C: 'I'm confident that the marble is red'. [Also consider versions with 'know' and 'think'.]
- (vii) (McGee): A: 'If a Republican wins, then if Reagan doesn't win, Anderson will win'; B: 'A Republican will win'; C: 'If Reagan doesn't win, Anderson will win'.
- (viii) (Better): We don't know yet who won. A: 'If a Republican won, then if Reagan didn't win, Anderson won'; B: 'A Republican won'; C: 'If Reagan didn't win, Anderson won'.

## 2. Two possible reactions to discuss today

- (a) "Syntactic mismatch": the trios aren't really of the relevant syntactic forms at all—i.e. despite appearances, B and C do not occur as *constituents* in A (at least on the natural parsing of A).
  - 1. *First version*: wide-scoping. E.g.: the logical form of 'If it's Tuesday we must be in Paris' is 'Must [[if it's Tuesday] [we be in Paris]]', or maybe 'We must [[be in Paris] [if it's Tuesday]]'. (Or something involving 'movement' which gets the same results in a somewhat more principled way.)
  - 2. *Second version*: Kratzerian syntax. E.g. the logical form of 'If it's Tuesday we must be in Paris' is '[Must [if it's Tuesday]] [we be in Paris]'.
- (b) "Semantic mismatch": certain key words that appear both in A and C—e.g. 'should', 'likely', the inner 'if' in the McGee example—are ambiguous (or context-sensitive),

and are not being interpreted in the same way when we judge A true and C false. If we fix the interpretation of these words in C to be the same one that's natural for A, then C is true.

### 3. More on syntactic mismatch

Challenge for wide-scopers: tell a story about the meaning of the embedded 'if' sentence and a story about the meaning of the modal (as a unary operator) that together yield plausible truth conditions for the A sentence.

Challenge for Kratzerians: tell a story about the meaning of the modal (as a binary operator) that yields plausible truth conditions for the A sentence.

Contrasts between the the wide-scoping response and the Kratzerian response.

- The wide-scoping response is particularly implausible in the case of the McGee example. Here, I suppose, it would go along with the thought that the real logical form of 'If a Republican won, then if Reagan didn't win Anderson won' is '[If Reagan didn't win] [[if a Republican won] [Anderson won]]'. But how could it be mandatory to reorder the 'if' clauses in this way? And if it's not mandatory, what could make it the preferred diambiguation in the circumstances?
- Kratzer says that her view allows "stacking" of 'if'-clauses, so that 'Modal if P, then if Q, then R' means the same as 'Modal if P and Q then R'. Presumably her favoured logical form is '[[Modal [if P]] [if Q]] R'. But she doesn't—at least in the paper we looked at—offer a story about how this actually comes about, compositionally.
- The following argument poses a special challenge for those want to solve the miners puzzle by wide-scoping:
  - (1) Should [if the miners are in shaft 1, we flood only one shaft]
  - (2) Should [if the miners are not in shaft 1, we flood only one shaft]
  - (3) Should [[if the miners are in shaft 1, we flood only one shaft] and [if the miners are not in shaft 1, we flood only one shaft]]
  - (4) Should [we flood only one shaft]
- While the miners puzzle is not so challenging for Kratzerian, it it does put pressure on Kratzer's own, Lewis-inspired, theory about the semantic values for modals (functions from propositions to functions from propositions to propositions).
  - On Kratzer's general theory, modals have two parameters of context-sensitivity: a *modal base* and an *ordering source*. We don't need to worry about what these are: what matters is that the modal base determines an *accessibility relation* R among possible worlds, and the ordering source determines a three-place relation  $w_1 \leq_{w_2} w_3$  among possible worlds.

- (Glossing over a certain complexity): given  $R$  and  $\leq$ , when  $M$  is a *necessity modal*,  $M(P)(Q)$  is true at  $w$  iff either no  $P$ -world bears  $R$  to  $w$ , or some  $P$ -world  $v$  bears  $R$  to  $w$  and is such that for every  $P$ -world  $u$  which bears  $R$  to  $w$  and is such that  $u \leq_w v$ ,  $u$  is a  $Q$ -world.
- Kratzer suggests that deontic modals like ‘should’ are necessity modals where the modal base is what’s “practically possible” and the ordering source is a matter of what’s best (by relevant standards). So  $\llbracket \text{Should} \rrbracket (P)(Q)$  is (roughly) true iff either  $P$  is practically impossible, or there’s a practically possible  $P$  world such that all practically possible  $P$ -worlds that are *at least as good as it* are  $Q$ -worlds. ]
- This validates the inference from (1) and (2) to (4) above, despite the misleading structure! No matter what the modal base and ordering source are for a necessity modal  $\Box$ , if  $\Box(P)(Q)$  and  $\Box(\neg P)(Q)$  are both true,  $\Box(\top)(Q)$  is also true.

#### 4. Similar examples involving ‘or’ rather than ‘if’

- (5) Either Bob is at home or he must be working
- (6) Either Bob is at home or he might be working
- (7) Either this is a zebra or it’s probably a donkey
- (8) Either we’re not going to leave tomorrow or we should start packing now

- Kratzerianism doesn’t help with these at all.
- Wide-scoping is much more syntactically implausible for these sentences. Also, wide-scoping seems to give wrong truth conditions for (6) and (7).
- Insofar as they seem to involve the same phenomenon as the earlier conditional cases, they tend to undermine the use of it to support any special view about conditionals (unless it goes along with some relevantly similar view about ‘or’).

#### 5. More on semantic mismatch

The proponent of semantic mismatch thinks that in the examples, the most natural readings for the embedded modals (and conditionals? and attitude verbs?) embedded in the consequents of conditionals is different from the most natural readings for the same words when unembedded.

Q1: is there a *possible* reading of the unembedded modal sentence that’s the same as the reading that’s natural for the consequent of the conditional?

Q2: is there a *possible* reading of the conditional where its consequent has the reading that’s natural for the unembedded modal sentence?

- If the answer to both questions was ‘no’, it’s unclear what we even mean by ‘valid’ for the relevant arguments. Any argument that doesn’t *have* any uniform interpretations will, vacuously, preserve truth on all uniform interpretations!

- Some evidence that the answer to Q2 is yes: '[Even] If John is robbing a bank, he shouldn't be robbing a bank'. '[Even] if John is [in fact] robbing a bank, he should be spending his life in Africa helping the poor'.
- Some evidence the answer to Q1 is yes: 'Either it's a donkey or it's likely to be a mule. There's a good chance that it's not a donkey. So, there's a good chance that it's likely to be a mule.'

*A view where the answer to both questions is 'yes':*

- (i) Model *questions* as *partitions of logical space*: sets of propositions (the *answers* to the question) such that exactly one of them is true at each world. Yes-no questions are two-cell partitions.
- (ii) At any point in a conversation certain questions are "under discussion". Utterance of a sentence generally puts under discussion both the yes-no question corresponding to the entire sentence, and the questions corresponding to its parts.
- (iii) In general, every modal and conditional is associated with an accessibility relation.
- (iv) Often (typically?), the accessibility relation for a modal or conditional uttered when certain questions  $Q_1, Q_2, \dots$  are under discussion is *constrained by*  $Q_1, Q_2, \dots$  — i.e. it's such that for  $v$  to be accessible from  $w$ ,  $v$  and  $w$  must belong to the same cell of  $Q_1$ , and belong to the same cell of  $Q_2$ , and...
  - E.g.: 'I'm pretty confident that this coin might land Tails', said of a coin chosen from a bucket with many ordinary coins and some double-headed coins. QUD: 'Is the coin double headed?'
  - When 'might' is constrained by the yes-no question 'A?', 'might P' is equivalent to '(A and might (A and P)) or (not-A and might (not-A and not P))'.
  - When 'if' is constrained by 'A?', 'if P, Q' is equivalent to '(A and if A and P, Q) or (not-A and if not-A and P, Q)'.
- (v) When we are interpreting B in 'If A, B', or 'A or B' or 'A and B', the yes-no question corresponding to A will typically be under discussion, and any modals that occur in B will be constrained by this question.
  - When this happens, 'If A, might P' will be equivalent to 'If A, (A and might (A and P)) or (not-A and might (not-A and not P))', which is in turn logically equivalent to 'If A, might (A and P)'.
  - 'If A, (if P, Q)' will be equivalent to 'If A, (A and if A and P, Q) or (not-A and if not-A and P, Q)', which is in turn logically equivalent to 'If A, (if A and P, Q)'.